observation is also essential, but, much of the time, experimentation is needed for a full understanding of those observations.

Three main types of ecological experiments are used to test theories: mathematical models, laboratory experiments, and field experiments. Each method has its merits, but it is only recently that the benefits of field experiments have been recognized. Before the 1960s, experiments outside a laboratory were a rarity.

A laboratory, however, is an artificial environment, where organisms may not behave as they do in their natural habitat. For example, bats leaving a roost at dusk may follow different routes to their foraging areas in spring and late summer. The potential reasons for the switch—changes in prey distribution and predator threats; seasonal differences in tree cover; or human disturbance and light pollution—cannot be established in a laboratory. Mathematical modeling might help predict patterns, but would be less effective at identifying the causes of change. To understand the bats' behavior, a study of their natural environment is crucial, and this is achieved only through research in the field.

Field experiments allow different factors to be manipulated to test their relevance. In the bat example, street lights could be switched off to evaluate the impact of light pollution on their behavior change.